Giacomo Galloni

Postdoctoral Researcher — Cosmology · University of Ferrara

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Research Interests

My research focuses on cosmic inflation through the Cosmic Microwave Background (CMB) and Gravitational Waves (GWs), with a methodological emphasis on likelihood analysis. I lead the large-scale CMB likelihood effort within the LiteBIRD collaboration and co-lead the Likelihood & Theory group of the Simons Observatory (SO), where I develop end-to-end pipelines that combine harmonic and pixel-based estimators with realistic component separation. In parallel, I am building a unified likelihood framework intended for use across current and future CMB experiments. On the science side, I have produced Bayesian and frequentist constraints on the amplitude and tilt of the primordial tensor spectrum, and I work on stochastic GW backgrounds and their cross-correlation with the CMB, including recent data-driven bounds on the quantum decoherence of inflationary GWs.

Positions & Education

Feb 2024 — present	**Postdoctoral Researcher**, University of Ferrara.
Nov 2020 — Jan 2024	**Ph.D. with honors in Astronomy, Astrophysics and Space Science**, University of Roma Tor Vergata. Supervisors: Prof. Nicola Vittorio, Prof. Sabino Matarrese.
Oct 2018 — Oct 2020	**M.Sc. with honors, Physics of the Universe**, University of Padova. Thesis: Prof. Nicola Bartolo, Dr. Angelo Ricciardone.
Oct 2014 — Oct 2018	**B.Sc. in Physics**, University of Padova. Thesis: Prof. Giacomo Ciani.

Fellowships, Visitings & Schools

Sep — Dec 2023 Funded visiting at IJCLab, Orsay (France).

Jun — Jul 2023 Funded visiting at Kavli IPMU, Tokyo (Japan) via MSCA CMB-INFLATE.

Nov 2020 — Jan 2024 Full Ph.D. fellowship, University of Roma La Sapienza.

Jun — Sep 2017 Funded internship at the University of Florida (bachelor thesis).

Mar 2022 GGI workshop on Astroparticle Physics, Cosmology and Gravitation, Firenze.

Dec 2021 Tonale Winter School in Cosmology, University of Heidelberg.

Nov 2021 Workshop on CLASS and Cobaya, University of Padova.

Selected Publications

Full list on Google Scholar and INSPIRE-HEP. First-author and small-group papers below.

Genesini, V., G. Galloni, et al. (2026). *"Cross-spectra likelihood for robust τ constraints from all satellite polarisation data"*. arXiv: 2603.22454 [astro-ph.CO]

Galloni, G., P. Campeti, et al. (2025). *"Accurate and efficient likelihood modeling for large-scale CMB data"*. JCAP 12, p. 052 — doi:10.1088/1475-7516/2025/12/052 — arXiv:2505.24829

Giardiello, S., A. J. Duivenvoorden, et al. (2025). *"Modeling beam chromaticity for high-resolution CMB analyses"*. Phys. Rev. D 111, p. 043502 — arXiv:2411.10124

Kruijf, J. de, G. Galloni, et al. (2025). *"The first data-driven bounds on the quantum decoherence of inflationary gravitational waves"*. arXiv: 2511.14727 [astro-ph.CO]

Galloni, G., S. Henrot-Versillé, et al. (2024). *"Robust constraints on tensor perturbations from cosmological data: a Bayesian and frequentist comparative analysis"*. Phys. Rev. D 110.6, p. 063511 — arXiv:2405.04455

Galloni, G., M. Ballardini, et al. (2023). *"Unraveling the CMB lack-of-correlation anomaly with the cosmological GW background"*. JCAP 10, p. 013 — arXiv:2305.18184

Galloni, G., N. Bartolo, et al. (2023). *"Updated constraints on amplitude and tilt of the tensor primordial spectrum"*. JCAP 04, p. 062 — arXiv:2208.00188

Galloni, G., N. Bartolo, et al. (2022). *"Test of the statistical isotropy of the universe using gravitational waves"*. JCAP 09, p. 046 — arXiv:2202.12858

Independent Projects

Ongoing	**CosmoForge — Power Spectrum and Likelihood Framework.** Python framework for power-spectrum estimation and likelihood analysis on large-scale CMB data, designed as a unified tool for LiteBIRD, SO, and future experiments.

In prep. **BICEP Diagnosis with the marginalized HL Likelihood.** Re-analysis of BICEP data exploiting new capabilities of the marginalized Hamimeche & Lewis (mHL) likelihood, stress-testing existing constraints on tensor modes.

Collaboration Projects

Jun 2024 — **LiteBIRD — Foreground and Sensitivity Studies.** Interfacing my Python package with foreground separation and sensitivity analyses.

May 2024 — **SO — Likelihood & Theory.** Co-lead of the group; maintenance and development of SOLikeT. Contributed single-mode likelihoods and led the restructuring of LAT MFLike.

Sep 2024 — **SO — Power Spectrum.** Contributing to the Power Spectrum group, with planned developments on QML-based estimators for large-scale modes.

Jan 2026 — **LiteBIRD — Moment-Expansion Component Separation.** Improving and characterizing the moment-expansion approach in the LiteBIRD context.

Jun 2023 — **LiteBIRD — MaPLEs team.** Leading efforts to study and characterize the likelihood of LiteBIRD; released within the collaboration a Python package to perform this analysis.

Apr 2023 — **LiteBIRD — Simulation Team.** Validation tests, cosmological analysis of CMB temperature with LiteBIRD, and development of parts of `litebird_sim`.

May 2022 — **LiteBIRD — Tests on Cosmic Inflation.** Monte-Carlo study of which inflationary models can be detected by LiteBIRD.

Service & Leadership

Sep 2024 — Co-lead, Likelihood & Theory group, Simons Observatory.

Dec 2025 — Early Career Coordination (ECC) representative, LiteBIRD.

Jan 2026 SOC member, LiteBIRD Face-to-Face Meeting.

Jul 2024 Session organizer & chair on the present/future of CMB, Marcel Grossman XVII (Pescara).

Selected Talks

May 2026 Invited Seminar at SISSA (Italy).

Apr 2026 Invited Speaker, GRaviCon 2026, Pisa (Italy).

Nov 2025 Invited Speaker, GW-CMB 2025, National Central University (Taiwan).

Sep 2025 Invited Speaker, Tensions in Cosmology, Corfu (Greece).

May 2025 Invited Outreach Talk, Pint of Science, Ferrara (Italy).

Dec 2024 Invited Seminar, University of Rome Tor Vergata (Italy).

Dec 2024 Invited Seminar, University of Padova (Italy).

Jul 2024 Invited Speaker, Marcel Grossman XVII, Pescara (Italy) — ICRANet.

Jul 2024 Invited Speaker, KEK CMB Group Meeting, Tsukuba (Japan).

Feb 2024 Invited Speaker, ASI-LiteBIRD, Frascati (Italy).

Dec 2023 Invited Speaker, Univers du pôle A2C, Orsay (France).

May 2023 Invited Seminar, University of Padova (Italy).

Oct 2022 Invited Speaker, LISA Community Call.

Apr 2022 Invited Seminar, University of Ferrara (Italy).

Mentorship

Feb 2026 — External mentor of a Ph.D. student at National Central University (Taiwan), working on r–n_t constraints from a joint analysis of LIGO-Virgo-KAGRA likelihoods and Simons Observatory forecasts.

Nov 2025 — Co-supervisor of a Master student at the University of Padova, on the recovery of r and n_t from SO forecasts.

Apr 2024 — Co-supervisor of a Ph.D. student at the University of Ferrara, who developed a cross-spectra likelihood for τ (Genesini et al. 2026), now working on moment-expansion component separation and reionization history.

Feb — Nov 2024 Supervisor of a Master student, Ferrara. *Thesis:* "Determination of the Reionization Optical Depth through a Combined Analysis of Planck and WMAP data."

Software Skills

Statistics. Expert in Bayesian and frequentist methods. Author of CosmoForge, a unified Python framework for QML and pixel-based CMB likelihood analyses; contributed to Cobaya; developed a profile-likelihood sampler.

Programming. Highly proficient in Python and C, with strong experience in package design, maintenance, and CI workflows. Author of CosmoForge, maintainer of SOLikeT, contributor to `litebird_sim` and several open-source cosmology repositories.

Boltzmann codes. Proficient user of CAMB and CLASS.